

8. LOCATING COMPONENT CENTERS OF GRAVITY

The purpose of this chapter is to provide guidelines for the determination of the location of centers of gravity for individual airplane components. Knowledge of component c.g. locations is essential in both Class I and Class II weight and balance analyses as discussed in Chapter 10 of Part II and Chapter 4 of this book.

In Part II, Chapter 10, Table 10.2 provides a summary of c.g. locations for the major structural components of the airplane only. In this chapter a slightly more extensive data base is provided. The presentation of component c.g. locations follows the weight breakdowns of Chapters 5-7:

- 8.1 C.G. Locations of Structural Components
- 8.2 C.G. Locations of Powerplant Components
- 8.3 C.G. Locations for Fixed Equipment

8.1 C.G. LOCATIONS OF STRUCTURAL COMPONENTS

Table 8.1 lists the most likely c.g. locations for major structural components. There is no substitute for common sense: if the preliminary structural arrangement of Part III (Step 19 of p.d. sequence 2, Part II) suggests that a given structural component has a different mass distribution than is commonly the case, an 'educated guess' must be made as to the effect on the c.g. of that component.

Example: Looking at the threeview of the GP-180 of Figure 3.47, p.86, Part II it is obvious that there is a concentration of primary structure at the aft end of the fuselage. The fuselage c.g. should therefore not be placed at 38-40 percent of the fuselage length, but probably at 55 to 60 percent.

8.2 C.G. LOCATIONS OF POWERPLANT COMPONENTS

Table 8.2 lists the most likely c.g. locations for powerplant components. Note that for engine c.g. locations manufacturers data should be used. 'Guessing' at engine c.g. locations is not recommended!

8.3 C.G. LOCATIONS OF FIXED EQUIPMENT

Table 8.3 lists guidelines for locating centers of gravity of fixed equipment components.

Table 8.1 Center of Gravity Location of Structural Components

Component:	Center of gravity location:
Wing (half):	<u>Unswept wing</u>: 38-42 percent chord from the L.E. at 40 percent of the semi-span. <u>Swept wing</u>: 70 percent of the distance between the front and rear spar behind the front spar at 35 percent of the semi-span
Horizontal Tail: (half)	Regardless of sweep angle: 42 percent chord from the L.E. at 38 percent of the semi-span.
Vertical Tail: (low tail)	Regardless of sweep angle: 42 percent chord from the L.E. at 38 percent vertical tail span from the root chord.
Vertical Tail: (T-tail) root	Regardless of sweep angle: 42 percent chord from the L.E. at 55 percent vertical tail span from the root chord.
Vertical Tail: (cruciform)	Regardless of sweep angle: 42 percent chord from the L.E. at between 38 and 55 percent vertical tail span from the root chord. Interpolate according to z_h/b_v .
Fuselage:	<u>Distances are given as a fraction of the fuselage length:</u>
Caution: Do not count the propeller spinner in fuselage or nacelle length!	Single engine tractors: 0.32-0.35 Single engine pusher: 0.45-0.48 Propeller driven twins: 0.38-0.40 (tractors on wing) Propeller driven twins: 0.50-0.53 (pushers on wing) Jet transports: 0.42-0.45 (wing mounted engines) Jet transports: 0.47-0.50 (rear fuselage mounted engines) Fighters: 0.45 (engines buried in the fuselage)
Tail booms:	0.40-0.45 of boom length starting from most forward structural attachment of the boom.
Nacelles:	0.40 of nacelle length from nacelle nose
Landing gear:	at 0.50 of strutlength for gears with mostly vertical struts

Table 8.2 Center of Gravity Location of Powerplant

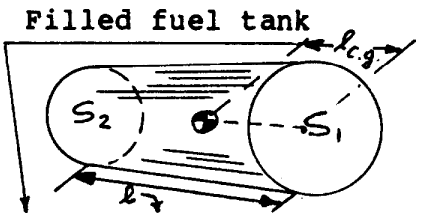
Components	
Component:	Center of Gravity Location:
Engine(s)	Use manufacturers data
Air induction system	Use the c.g. of the gross shell area of the inlets
Propellers	On the spin axis, in the propeller spin plane
Fuel system	Refer to the fuel system layout diagram required as part of Step 17 in p.d. sequence II, Part II, p.18.
Filled fuel tank	Assuming a prismatic shape (See figure left), the c.g. is located relative to plane S_1 at:
 $l_{cg} = (1/4) \{S_1 + 3S_2 + 2(S_1 S_2)^{1/2}\} / \{S_1 + S_2 + (S_1 S_2)^{1/2}\}$ (8.1)	
Trapped fuel and oil	Trapped fuel is normally located at the bottom of fuel tanks and fuel lines. Trapped oil is normally located close to the engine case.
Propulsion system	Make a list of which items contribute to the propulsion system weight and 'gueseimate' their c.g. location by referring to the powerplant installation drawing required in Step 5.10, pages 133 and 134 in Part II.

Table 8.3 Center of Gravity Location of Fixed Equipment
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Component:	Center of Gravity Location:
Flight Control System	<u>Note:</u> for all systems, the c.g. location can be most closely 'guestimated' by referring to the system lay-out diagrams described in Part IV of this text. These system lay-outs were required as part of Step 17 in p.d. sequence 2, Part II, p.18.
Hydraulic and Pneumatic System	
Electrical System	
Instrumentation, Avionics and Electronics	
Air-conditioning, Pressurization, Anti-icing and de-icing System	
Oxygen System	
Auxiliary Power Unit	See engine manufacturer data.
Furnishings	Refer to the fuselage internal arrangement drawing required by Steps 4.1 and 4.2 in Part II, pp 107 and 108.
Baggage and Cargo Handling Equipment	
Operational items	See furnishings
Armament	This item is normally close to the cockpit
Guns, launchers and weapons provisions	From manufacturer data.
Flight test instrumentation	A sketch depicting the locations of sensors, recorders operating systems should help in locating the overall c.g. of this item.
Auxiliary gear	Make a list of items in this category and 'guestimate' their c.g. locations.
Ballast	Ballast weights are normally made from lead. Ballast c.g. is thus easily located.
Paint	Centroid of painted areas.

9. CLASS II WEIGHT AND BALANCE ANALYSIS

The basic method used in performing a Class II weight and balance analysis is identical to that used for the Class I weight and balance analysis. The latter was discussed in detail in Part II, Chapter 10. The only difference is, that a more detailed weight statement is used: the Class II weight prediction method of Chapters 4-7 is used.

During this stage of the preliminary design frequent questions which are raised, are:

1. How much does the overall airplane c.g. move as a result of moving some component?
2. How much does the airplane static margin change as a result of moving the wing?

These questions are answered in Sections 9.1 and 9.2 respectively.

9.1 EFFECT OF MOVING COMPONENTS ON OVERALL AIRPLANE CENTER OF GRAVITY

Figure 9.1 illustrates an airplane, its c.g. location and the c.g. location of component i. The overall center of gravity of the airplane is found from:

$$x_{cg} = \frac{\sum_{i=1}^{i=n} W_i x_i}{\sum W_i} \quad (9.1)$$

Evidently:

$$\sum_{i=1}^{i=n} W_i = W \quad (9.2)$$

The rate at which overall airplane c.g. moves, when a component i is moved, can be found by differentiation of Eqn. (9.1):

$$\partial x_{cg} / \partial x_i = \frac{W_i}{\sum_{i=1}^{i=n} W_i} \quad = (9.3)$$

If component i is moved over a distance Δx_i , the

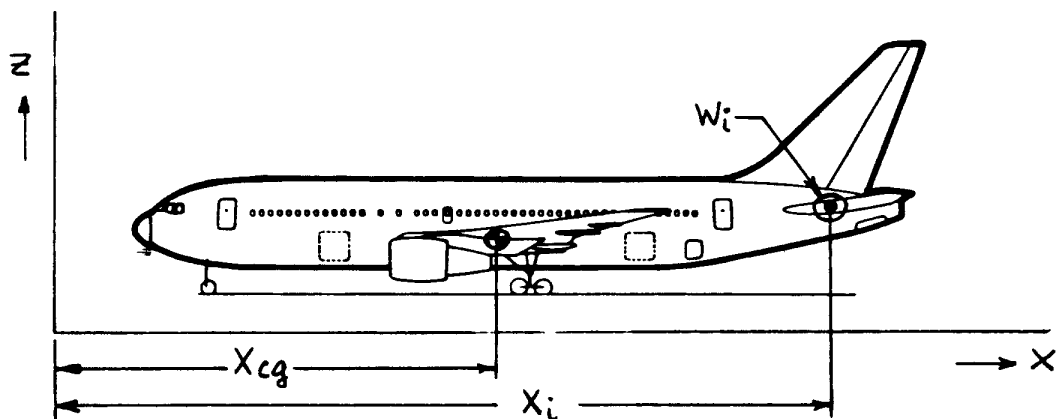


Figure 9.1 Definition of Overall C.G. Location and of Component C.G. Location

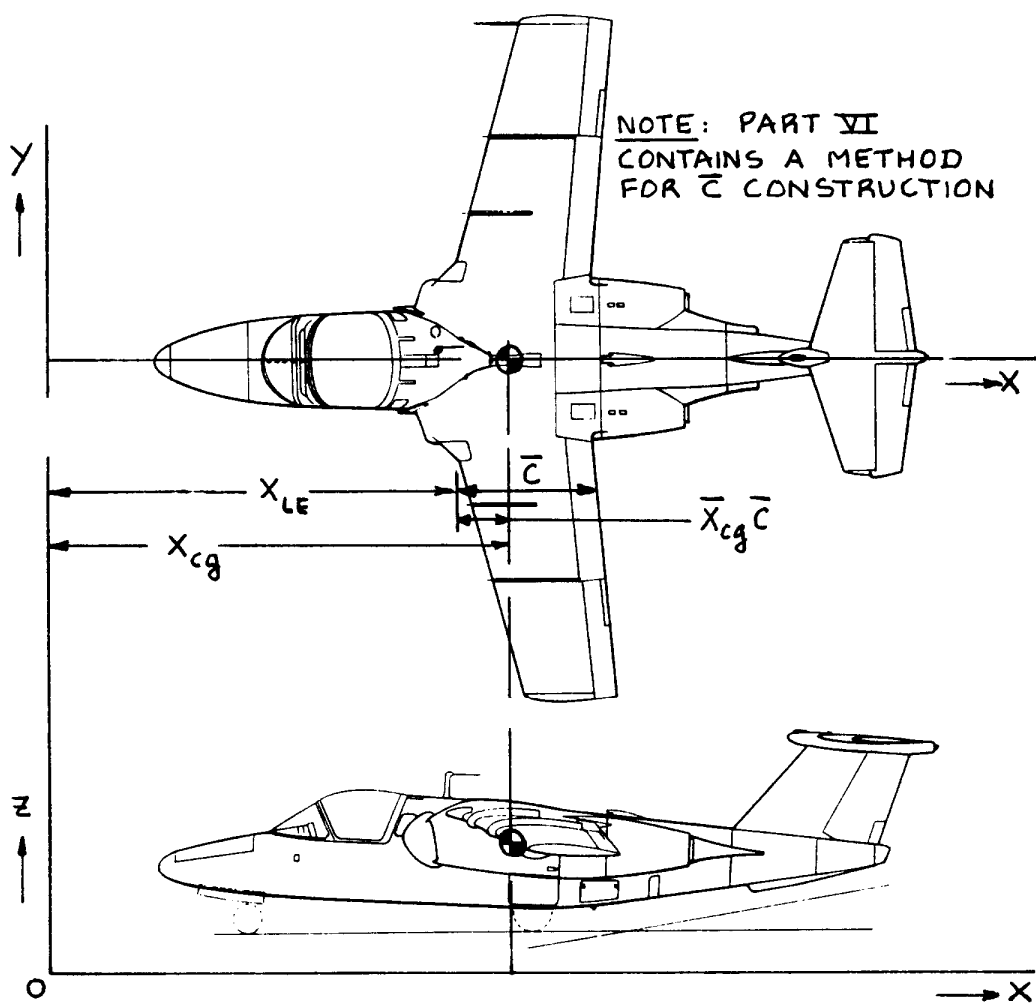


Figure 9.2 Definition of Mean Geometric Chord ($\bar{c}$) Location and Overall C.G. Location

overall airplane c.g. moves over a distance given by:

$$\Delta x_{cg} = (\Delta x_i)(W_i) / (\sum_{i=1}^{i=n} W_i) \quad (9.4)$$

Equation (9.4) suggests that to move the overall c.g. of the airplane significantly, either a heavy weight component can be moved a small distance or a light weight component can be moved a large distance.

Items which are frequently moved about to achieve satisfactory weight and balance results are: batteries, air-conditioner units, certain 'black' boxes and sometimes just plain ballast. The reader will note from the detailed weight statements in Appendix A that several airplanes carry a relatively large amount of ballast.

9.2 EFFECT OF MOVING THE WING ON OVERALL AIRPLANE CENTER OF GRAVITY AND ON OVERALL AIRPLANE AERODYNAMIC CENTER

Figure 9.2 illustrates an airplane, its mean geometric chord location and its overall c.g. location.

If the leading edge of the mgc of the wing is at fuselage station (FS) x_{LE} , the airplane c.g. in terms of

the wing mgc can be written as:

$$\bar{x}_{cg} = (x_{cg} - x_{LE}) / \bar{c} \quad (9.5)$$

When a component i is moved over a distance Δx_i ,

the overall airplane c.g. moves relative to the wing mgc as:

$$\Delta \bar{x}_{cg} = (\Delta x_i)(W_i) / (\bar{c} \sum_{i=1}^{i=n} W_i) \quad (9.6)$$

For a conventional, tail-aft airplane, its aerodynamic center location can be written as:

$$\bar{x}_{ac} = \{C_1 + C_2(\bar{x}_{ac_h})\} / (1 + C_2) \quad (9.7)$$

$$\text{where: } C_1 = \bar{x}_{ac_w} + \Delta \bar{x}_{ac_{wb}} \quad (9.8)$$

$$C_2 = (C_{L_{\alpha_h}} / C_{L_{\alpha_{wb}}}) (1 - ds/da) (S_h/S) \quad (9.9)$$

A detailed derivation of Eqn. (9.7) may be found in Reference 19, p.133.

Part VI contains methods for computing the liftcurve slopes and aerodynamic centers which appear in C_1 and in C_2 .

Warning: the wing-body aerodynamic center shift, $\Delta \bar{x}_{ac_{wb}}$ in Eqn.(9.8) is always a negative number: it shifts the a.c. forward!

If the wing is moved aft over a distance Δx_w , the overall airplane c.g. is:

$$\bar{x}_{cg_{new}} = \bar{x}_{cg_{old}} + (\Delta x_w W_w) / (\bar{c} \sum_{i=1}^{i=n} W_i) \quad (9.10)$$

The new a.c. location can now be written as:

$$\bar{x}_{ac_{new}} = \{C_1 + C_2 (\bar{x}_{ac_h} - \Delta x_w / \bar{c})\} / (1 + C_2) \quad (9.11)$$

Equations (9.10) and (9.11) can be used to 'redo' the X-plot of Part II, Chapter 11. This 'redone' X-plot in turn is used to:

1. determine how much the horizontal tail area must be changed as a result of moving the wing

or:

2. determine which other weight components need to be moved and by how much, to maintain some desired level of stability (or instability as the case may be).

For a canard airplane and/or for a three surface airplane similar equations are easily derived. The reader should consult Equations (11.1) and (11.2) in Part II for guidance.